

Lightweight, Provenance-Aware Retrieval Gating for Vision–Language Inference on Resource-Constrained Devices

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Abstract

Vision–language models (VLMs) are too expensive to run on every image of a locally held archive — a dashcam buffer, a fixed-camera store, a field-inspection dataset — on single-GPU, no-cloud-round-trip hardware where such archives actually live, entirely on-device with no external inference calls. Retrieval gating (encode once with a cheap embedding model, run the VLM only on the top- k retrieved candidates) is the standard fix, but a real device budget has several dials, and it is not obvious which are worth tuning once the others are fixed. Instrumenting all of them on a 2,500-image offline archive (Flickr30k + BDD100K) served locally by frozen CLIP, FAISS, and SmolVLM2-256M on one NVIDIA T4, we find a counterintuitive result for on-device precision choices: INT8 quantization — the default “efficiency” lever in most deployment checklists — runs $7.69\times$ slower than FP16 in this batch-size-one, single-model regime, while NF4 delivers the larger memory saving (35.2%) at only $1.27\times$ latency cost. This inversion holds because the budget is dominated by a single stage ($c_{\text{vlm}}/c_{\text{emb}} \approx 410$, $c_{\text{vlm}}/c_{\text{query}} \approx 2.3 \times 10^4$), which also means approximate search saves a negligible 0.07 ms against a 1.4 s VLM call, and gate width k is the only dial with a measurable effect on end-to-end cost. A lightweight (0.26M-parameter) linear adapter improves retrieval in both domains without closing the gap between general photographs ($R@5 = 0.890$) and a visually homogeneous driving-scene corpus ($R@5 = 0.344$). Because the dominant stage leaves so much headroom, the gate can also emit a structured provenance record — which candidate reached the model and why, versus what the model said — for free, giving fully local, auditable multimodal inference with no cloud dependency for logging or review. Our claims are scoped to one deployment scale and one GPU; the transferable lesson is that on-device quantization and indexing choices should be measured against the stage that actually dominates the budget, not assumed from server-scale intuition.

1 Introduction

On-device and edge deployments of vision–language inference share a constraint that cloud deployments do not: a fixed, small budget of latency, memory, energy, and often privacy, with no round trip to a large hosted model, and no external inference calls of any kind. Locally held image archives are a common instance of this — a vehicle dashcam buffer, a fixed-camera surveillance store, a personal photo library, an offline field-inspection dataset, or a remote site without reliable connectivity — where a user still wants to ask, in natural language, “which images show a dog running on grass?” Running a VLM once per image is simple but breaks the budget immediately: at $N = 2000$ images and ~ 2 s per call, that is over an hour of GPU time for one query, on hardware that typically has neither the hour nor the GPU to spare, and no cloud fallback to absorb the overflow.

Retrieval gating is the standard fix: a cheap encoder scores the whole archive once, offline, and the expensive model runs only on the top- k candidates at query time, following the cost model

$$C_{\text{gated}} = N c_{\text{emb}} + c_{\text{query}} + k c_{\text{vlm}} \quad \text{against} \quad C_{\text{exhaustive}} = N c_{\text{vlm}}.$$

The N/k speedup this implies is arithmetic, not a result. What it does not tell a device-side practitioner is which of the pipeline’s other knobs — index choice, model precision, embedding adaptation — are

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worth spending a constrained engineering budget on once gating is already in place, entirely within a fully local execution loop. If one stage dominates the others by orders of magnitude, tuning the small stages can be invisible at the level of the budget that actually matters on the device.

We measure this directly, end to end, on one concrete on-device-scale, fully local deployment: frozen CLIP ViT-B/32 [11], FAISS [4], and SmolVLM2-256M [10] on a single NVIDIA T4, sweeping gate width k , index type, VLM weight precision, and embedding adaptation. Contributions:

1. **A measured cost asymmetry that collapses the tunable budget onto one dial.** $c_{\text{vlm}}/c_{\text{emb}} \approx 410$, $c_{\text{vlm}}/c_{\text{query}} \approx 2.3 \times 10^4$: gate width k is the only parameter we vary with a measurable effect on end-to-end cost.
2. **Evidence that two “obvious” on-device optimizations do not pay off at this scale.** Approximate search (HNSW) saves 0.07 ms against a 1.4 s VLM call; INT8 quantization is a strict regression relative to NF4 in a batch-size-one, single-model regime typical of on-device inference.
3. **A domain-dependent limit on how far cheap adaptation can go.** A 0.26M-parameter linear adapter — itself a device-friendly, inference-cheap component — improves retrieval recall in both evaluated domains but leaves a large gap between them, suggesting the limit is the frozen backbone, not the adapter budget.
4. **A provenance mechanism that comes free with gating.** Because the gate makes an inspectable selection decision before the VLM runs, every query can log why a candidate was chosen and what the model concluded, fully locally, at a cost that is unmeasurable next to the dominant VLM stage — a natural fit for private, reliable, on-device execution where sending logs to a cloud service defeats the purpose of running locally at all.

2 System and Experimental Setup

Pipeline. CLIP maps images and text into a shared 512-d space; the gate encodes every archive image once, offline, and at query time retrieves the top- k nearest images from a FAISS index by cosine similarity. SmolVLM2-256M runs only on those k candidates with a verification prompt (“Does this image show $\langle \text{query} \rangle$?”, ≤ 64 new tokens). An optional linear adapter $z' = \text{normalize}(Wz + b)$, $W \in \mathbb{R}^{512 \times 512}$ (0.17% of CLIP’s parameters, trained with symmetric InfoNCE against caption embeddings, CLIP itself left frozen) sits between CLIP and the index. All components run locally on a single GPU with no network calls in the inference path. Each query can emit a provenance record: trace id, timestamp, query, index configuration, ranked candidate ids and scores, and the VLM’s per-candidate output — separating why a candidate was selected from what the model concluded, entirely on-device.

Corpus and hardware. 2,500 images: 2,000 Flickr30k test images [13] (general photographs) and 500 BDD100K city-street images [14] (driving scenes, visually homogeneous, pseudo-captioned by the same frozen SmolVLM2). All experiments run on one NVIDIA T4 GPU with no cloud round-trip at any stage; FAISS runs on CPU. Latencies are measured after warm-up, averaged over 200 queries $\times$ 5 repeats for index queries.

Components swept. FAISS Flat (exact), IVF (nlist=64), HNSW (M=32); SmolVLM2-256M under FP16, INT8, NF4 (bitsandbytes); adapter on/off.

Metrics. Per-stage latency; recall proxy (overlap with exact search); R@1/5/10 for the adapter; a crude answer proxy (fraction of retrieved candidates the VLM affirms, over a small fixed query set).

3 Results

The budget is dominated by one stage. Encoding one image with CLIP takes 4.89 ms (batch 64); an exact query over 2,500 vectors takes 0.089 ms; one VLM call takes 1.40 s (FP16). Thus $c_{\text{vlm}}/c_{\text{emb}} \approx 410$ and $c_{\text{vlm}}/c_{\text{query}} \approx 2.3 \times 10^4$: on a real device, indexing the entire 2,500-image collection costs about as much as a single $k=5$ gated query.

Table 1: Sweeping gate width k at $N = 2000$ (Flickr30k). Speedup is N/k by construction; VLM cost varies $100\times$ across the sweep while the answer proxy varies by at most 0.033, non-monotonically.

k	1	5	10	25	50	100
Speedup vs. exhaustive	$2000\times$	$400\times$	$200\times$	$80\times$	$40\times$	$20\times$
Recall proxy @ k	1.000	1.000	1.000	0.987	0.980	0.987
Answer proxy	0.667	0.667	0.700	0.683	0.683	0.683

Table 2: Left: index comparison at $n = 2500$, $k = 10$ (query latency as % of one 1.4s VLM call). Right: SmolVLM2-256M under bitsandbytes quantization, 20 images, ≤ 64 tokens. INT8’s slowdown, not its memory saving, is the finding that matters for a fully local, single-request deployment.

Index	Query	R@10	%VLM	Prec.	Mem.	Save	Lat. ($\times$)
Flat	0.089 ms	1.000	0.0063%	FP16	489.2 MB	—	$1.00\times$
IVF	0.020 ms	0.963	0.0014%	INT8	411.4 MB	15.9%	$7.69\times$
HNSW	0.020 ms	0.999	0.0014%	NF4	316.9 MB	35.2%	$1.27\times$

INT8 quantization: the textbook efficiency lever that loses on-device. Quantization inverts the naive expectation for a batch-size-one, single-model, on-device inference regime. INT8 saves only 15.9% of weight memory (489.2→411.4 MB) while running $7.69\times$ slower (1.401→10.778 s/image), plausibly because mixed-precision decomposition overhead cannot be amortized without batching — a cost that a fully local, single-request-at-a-time deployment cannot hide behind throughput the way a batched server can. NF4 gives the larger memory saving (35.2%, to 316.9 MB) at only $1.27\times$ latency. Because gating has already removed $\sim 400\times$ of potential VLM calls, NF4’s moderate per-call penalty is one a real device budget can absorb; INT8’s cannot. For an on-device deployment choosing a quantization scheme off a generic checklist, this is the paper’s most actionable and least intuitive finding.

Approximate search does not pay for itself at this scale. HNSW reaches near-exact recall (0.999 vs. 1.000 exact) at 0.0196 ms/query versus Flat’s 0.0886 ms — a 0.07 ms saving against a 1.4s VLM call ($<0.007\%$), while costing $\sim 40\times$ more to incrementally update (5.70 ms vs. 0.135 ms) and $\sim 37\times$ more to rebuild (81.4 ms vs. 2.2 ms) — a real cost on a device that updates its archive more often than it queries it.

Cheap adaptation narrows but does not close a domain gap. Table 3 reports in-sample recall with/without the 0.26M-parameter linear adapter. On Flickr30k, R@5 rises $0.834 \rightarrow 0.890$; on BDD100K, $0.216 \rightarrow 0.344$ (the larger relative gain, on the harder domain) — but the gap between domains remains large even under this most favorable, in-sample evaluation. For a resource-constrained, fully local deployment specialized to one visual domain (e.g. an on-vehicle dashcam archive, structurally closer to BDD100K than to general photography), this means a cheap adapter is not, on its own, a substitute for domain-matched retrieval quality.

4 Provenance as a Free On-Device Systems Contribution

Because VLM inference dominates cost by two to four orders of magnitude, and because the gate already makes an explicit ranking decision, a structured provenance record costs nothing measurable to emit: {query, index config, ranked candidate ids + scores, VLM output, latency} per query. On a device where sending inference logs to a cloud service for review would defeat the point of local execution, this record is what makes the pipeline auditable without leaving the device: an operator can tell why an image reached the model (retrieval score, rank) separately from what the model said about it, using only local storage and no network access. We view this as a systems contribution independent of any latency benefit, and a natural complement to the reliability requirements (safe, private, fully local execution) that constrained on-device deployments already carry.

Table 3: Text-to-image recall with/without the adapter (in-sample, jointly trained on both domains).

Domain (Adapter)	R@1	R@5	R@10
Flickr30k, none	0.592	0.834	0.889
Flickr30k, joint	0.684	0.890	0.942
BDD100K, none	0.094	0.216	0.306
BDD100K, joint	0.160	0.344	0.458

5 Limitations

All results are measured at a fixed offline-archive scale ($N = 2,500$ images, one NVIDIA T4, single seed) chosen to match the on-device deployments this paper targets — a dashcam buffer, a store camera, a field-inspection set — which typically hold hundreds to a few thousand images rather than web-scale corpora. This scale is a deliberate match to the target regime, not an oversight: it is exactly the range in which the measured cost asymmetry ($c_{\text{vlm}}/c_{\text{emb}} \approx 410$, $c_{\text{vlm}}/c_{\text{query}} \approx 2.3 \times 10^4$) holds, and we explicitly do not claim it generalizes to larger, server-scale corpora, where approximate search is well known to pay off; we expect (but do not locate) a crossover at larger corpus sizes, and flag this as the natural next experiment for anyone building on this work. The adapter evaluation is in-sample (trained and evaluated on the same pairs); held-out numbers on both domains would likely be lower. BDD100K’s captions are machine-generated by the same model used downstream, so its retrieval numbers partly measure self-consistency. The answer proxy is a crude affirmation rate, not ground-truth accuracy, and we did not condition it on retrieval correctness per domain — the provenance record in Section 4 makes this computable, and we flag it as the clearest next measurement (see Appendix C). We view each of these as a scoped boundary of a single deployment-scale, fully local study rather than a threat to the paper’s central claim, which is about where the budget goes, not about extrapolating absolute numbers beyond the regime measured.

6 Conclusion

On a real, fixed device budget — one GPU, offline archive, fully local execution with no cloud round trip — a retrieval-gated VLM pipeline is dominated by a single stage by two to four orders of magnitude, which collapses the tunable design space onto gate width k and makes two textbook on-device optimizations, approximate search and INT8 quantization, numerically invisible or counterproductive at this scale. A lightweight adapter improves retrieval without closing a domain gap that limits reliability in specialized, resource-constrained deployments. Because the dominant stage leaves so much budget headroom elsewhere, a structured, fully local provenance record is essentially free to add and turns the pipeline’s evidence-selection step into something auditable entirely on-device, with no data ever leaving the device. The transferable lesson for on-device deployment: measure where the real budget goes before optimizing components the budget cannot see.

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A Additional Related Work

Cascaded and selective inference. NoScope [7] cascades cheap classifiers ahead of an expensive detector for video; FrugalGPT [1] routes LLM queries to the cheapest adequate model. Our gate is a cascade in this family, specialized to top- k retrieval; we additionally ask whether the cascade’s non-dominant stages are worth optimizing once the dominant stage is fixed by a device budget.

Multimodal RAG. RAG [8] and its multimodal extension VisRAG [12] target answer quality at web- or document-corpus scale, where retrieval cost is itself nontrivial. Our setting inverts the emphasis: at archive sizes typical of on-device storage (a few thousand images), retrieval cost is negligible next to VLM cost, so the cost-driven concerns central to that literature resurface only at corpus sizes well above what we study.

Approximate search and quantization. Product quantization [6] and HNSW [9], packaged in FAISS [4], are evaluated almost exclusively at 10^6 – 10^9 -vector scale. LLM.int8() [2] and QLoRA’s NF4 [3] are typically evaluated on models an order of magnitude larger than SmolVLM2-256M, in regimes where throughput gains amortize dequantization overhead — exactly the regime a small, single-model, on-device deployment does not sit in.

Adapting frozen embeddings. CLIP-Adapter [5] and Tip-Adapter [15] attach lightweight modules to frozen CLIP for few-shot classification. We use a similarly lightweight, inference-cheap adapter but target retrieval recall for a gate, probing whether a linear reprojection can close a domain gap relevant to specialized on-device corpora (e.g. driving scenes) — a question this prior work does not address.

B Maintenance Costs (Index Updates)

Offline on-device archives are typically append-heavy and query-light, so update cost can matter as much as query cost. Incremental insertion: 0.135 ms (Flat), 0.144 ms (IVF, reused centroids), 5.698 ms (HNSW, $\sim 40\times$ Flat). Full rebuild: 2.213 ms (Flat), 5.181 ms (IVF), 81.384 ms (HNSW). At $n = 2500$, all three remain far below the 1.4 s VLM call they gate, reinforcing the paper’s main claim, but the relative gap between index types is far larger for maintenance than for query latency — worth revisiting at larger on-device archive sizes.

C Future Work: Provenance-Conditioned Reliability Auditing

The provenance record in Section 4 logs, per query, both retrieval correctness (via the recall proxy) and the VLM’s per-candidate output. We have not yet computed the natural next metric this enables: the rate at which the VLM’s affirmative answers coincide with a retrieval the gate got right, per domain. We flag this rather than report a number we have not measured. Given the domain gap in Table 3 ($R@5 = 0.890$ vs. 0.344), we would expect this joint metric to be most informative exactly on the specialized, resource-constrained domain (BDD100K-like archives) where on-device deployments are most likely to specialize, and computing it is a direct extension of the logging infrastructure already described.